

Chain Link Security: Trends in Simultaneous HTTPS RR, ECH, and DNSSEC Use

Yu-Siang Chou
College of Engineering
Oregon State University
Corvallis, OR
chouyu@oregonstate.edu

Richard Mwaiwo Ngeti
College of Engineering
Oregon State University
Corvallis, OR
ngetir@oregonstate.edu

Alexander Yang Rhoads
College of Engineering
Oregon State University
Corvallis, OR
rhoadsal@oregonstate.edu

Abstract—DNS HTTPS RR, ECH, and DNSSEC are relatively new, and potentially mutually beneficial, parts of web security infrastructure. In order to discover the wider environment for the deployment of this protocol suite, we collected and extracted the DNS HTTPS record data of the top 10,000 domains, calculated the share of these protocols among those domains, and then split the data into another list of domains that carried each protocol by itself and combined with others. In turn, the list containing the domains that use all three protocols was examined for common categorical factors among the clients of the suite. The results demonstrate the low usage for ECH, DNSSEC, and the combined use of the record with the two protocols, among that top set of domains. We examine the reasons for this, finding that it could be caused by the components' facilities, complexity of deployment and maintenance, or limitations. We also examine the incentives that the domains using the entire security set, finding why a domain would want to use them, despite the difficulties.

I. INTRODUCTION

HTTPS Resource Record is a DNS resource record used to configure HTTPS, as well as optimize its usage. Published in November 2023, though existing earlier in draft form, the records allow a domain to perform a number of useful actions, like mitigate preloading processes and utilize the Encrypted Client Hello (ECH) protocol to allow for stronger privacy protections.

Notably, the non-standardized ECH is incorporated into these records, which potentially enables further adoption of the protocol. Encrypted Client Hello allows for the encryption of TLS handshakes, which enables a client to better hide their domain information.

Older security protocols, like the Domain Name System Security Extensions (DNSSEC) suite, can be used to further enhance the security of the integrated HTTPS RR and ECH suite. DNSSEC specifically authenticates DNS records to prevent spoofing attacks and ensure data integrity. Theoretically, HTTPS RR can utilize DNSSEC to fortify its own domain data, whilst using its ECH functionality to enhance the privacy of its clients, whose cryptographic keys are secured under the additional protection of DNSSEC.

Despite that promise, previous research found that there was a low record of simultaneous deployment of HTTPS RR with ECH and HTTPS RR with DNSSEC in 2024, with no information on the deployment of the record type simultaneously with the aforementioned protocols [1]. Since

HTTPS RR is a relatively new record protocol, it is worth keeping up with the broader trends in its deployment and support into 2026, as well as scrutinizing the reasons for those trends.

In this paper, our goal is to discover the larger environment for the deployment of DNS HTTPS resource records with ECH and DNSSEC in 2026, within the specific context of the top 10,000 domains. Exploring the top 10,000 domains is a different beast than exploring the higher number of domains that tend to be examined in other studies. It describes a smaller top percentage, and thus overall more popular, set of domains that are more likely to have the resources to support these records and protocols. It can be thought of as a case study for the adoption of HTTPS RR, ECH, and DNSSEC among more popular domains. This will be done by examining the current environmental trends in HTTPS RR, ECH, and DNSSEC usage in terms of deployment numbers, the different domains that use them, the larger incentives domains have in using the technology, and the current hurdles in managing the usage of all simultaneously. The data will be provided both by extracting it directly from popular domains and by qualitatively categorizing the extracted data by hand.

II. EXISTING WORK

Prior 2024 research on the HTTPS RR environment studied the top 1 million domains for each month examined [1]. It studied these domains over different varying periods of time, depending on the research area examined, from 2023 to 2024. The 11-month period from June 2023 to April 2024 saw a monthly range of HTTPS RR adoption of 20% to 27% of domains. ECH deployment was studied, but exact numbers were not provided, due to Cloudflare, a DNS provider that acts as a major source of ECH deployment, turning it off from October 2023 to August 2024, due to unknown technical issues. One aspect of ECH that was noted was its difficulty of management, as any domain that uses it will have to maintain the rollover of encryption keys, which was set to occur every 1 to 2 hours in most observed domains. Key rollover is something both clients and servers have to manage; otherwise, they face connection failures. DNSSEC usage over that aforementioned 11-month period of time saw roughly 7% of HTTPS RR apex domains use the protocol suite. The

HTTPS RR parameters surveyed were complex, as domains with Cloudflare had an enforced default parameter setting that was difficult to turn off, with 79% of Cloudflare domains using the default, and no tracking on non-default domains' parameter settings. The default had Service mode turned on, usage of alpn, ipv4hint, and ipv6hint. Other DNS providers had around a 95% usage rate for all parameters, with the exception of SvcPriority. That depended on the particular provider, but if that provider favored ServiceMode or AliasMode, usage for the particular mode would be in the 95% range.

Earlier research found a similar trend in parameter usage, with an overwhelming majority of the 10.56 million HTTPS records surveyed using alpn, ipv4hint, and ipv6hint at a roughly 95% share or higher in 2023 [2]. This same survey also found that ServiceMode was used in a similar share of domains, taking 10.55 million to AliasMode's 2,600. ECH usage, however, was found in only 20 HTTPS resource records. This is a notable early step in HTTPS RR adoption, as 2023 was the year it was officially launched.

III. BACKGROUND

In this section, we will discuss the relevant background on HTTPS & TLS, Encrypted Client Hello (ECH), the Domain Name System (DNS), Domain Name System Security Extensions (DNSSEC), and the HTTPS RR parameters.

A. HTTPS & TLS

HTTPS is used to securely communicate between the client and the server for services that require them [3]. It uses TLS (Transport Layer Security) as a cryptographic protocol to provide encryption and authentication for HTTP protocols during the TLS handshake.

B. HTTPS Resource Record (HTTPS RR)

HTTPS Resource Records, or HTTPS RR, are a type of DNS record meant to deliver configuration and parameters for how to access a particular service via HTTPS. It's used to optimize the process of connecting to a domain using HTTPS, and tells browsers that all useful information is still reachable when upgrading the connection from HTTP [4].

Not every HTTPS domain uses HTTPS RR, in fact the majority don't. They were officially defined and published in November of 2023, though saw prior usage while being drafted. On November 13 2023, within the month of its publishing, it was found that, out of around 10 million domains, HTTPS RR usage was 4% for bare domains, and 4.4% for domains that use the www service name. That number increases to 25.6% for bare domains, and 25.5% for www domains when examining Tranco's list of the top 1 million domains [5].

C. Encrypted Client Hello(ECH)

ECH is an extension of TLS that hides the identity of the target TLS server by encrypting the Server Name Indication (SNI) in the Client Hello using the public key of the client-facing server, which limits access to the particular user visiting a domain [6].

D. Domain Name System Security Extensions(DNSSEC)

The Domain Name System Security Extension (DNSSEC) is an authentication protocol suite. The DNSSEC resolver is used to verify that information originates from a legitimate server, and that data matches the information on authoritative DNS servers by checking the signatures in the DNSSEC servers' answers. This process helps protect against a number of DNS vulnerabilities such as DNS Spoofing, Cache Poisoning, Man-in-the-Middle Attacks, and the lack of Integrity Checks [7].

A noted flaw of the suite is that the cryptographic key management is difficult, slowing DNSSEC deployment on a global scale [8].

E. HTTPS RR Parameters

These are the series of parameters that an HTTPS RR can use to configure HTTPS. SvcPriority refers to the priority of the record. A value of 0 indicates AliasMode, aliasing a domain to the target domain, and other values indicate ServiceMode, which provides information specific to a service endpoint. TargetName is a domain name, which can be either the alias target in AliasMode or the alternative endpoint in ServiceMode. SvcParams, utilized only in ServiceMode, is a list of key-value pairs that are included to provide details about the endpoint. The current specification defines seven parameter keys: port, alpn, no-default-alpn, ipv4hint, ipv6hint, and mandatory. Of what's covered in the paper alpn (Application-Layer Protocol Negotiation) specifies additional application protocols. Ipv4hint/ipv6hint suggests ipv4/ipv6 addresses for reaching the endpoint [9].

IV. METHODOLOGY

Our approach can be sectioned off into two steps. The first step was to fetch data from the domains themselves. This was done with a list requested from the site Tranco, which averages the popularity ranking of domains from several source lists, weighting them with the Dowdall rule [10]. What was specifically requested was a list that gave the ranking of the top 10,000 domains from January 1, 2025, to December 31, 2025. It was configured to give a list that only included pay-level domains and to only include ones not flagged as dangerous by Google Safe Browsing. The final list we used was requested on March 13, 2026, and had its data extracted the same day. While the list itself describes the most visited domains throughout 2025, the data extracted from these domains is reflective of these domains at the date of extraction. That means all analyses done on the data reflect the state of domains on March 13, 2026.

The second step was to make a qualitative analysis of the domains fetched. Self-created lists are created and examined, made by checking the raw data extracted from each domain, then dividing them into smaller lists based on the presence of different factors. The specific list we examined was of the domains that used HTTPS RR + ECH + DNSSEC. This was done so by hand, where each domain is assigned a series of categories based on what type of domain it is. The

categories were also self-made, and thus subject to the biases of our research group, and are codified in the qualitative data collection chapter of this section.

A. Domain Data Extraction Approach

Data was extracted from the list by reading through the set of domains, then trying to extract specific bits of data by both trying to make a TLS handshake with the domain to check for HTTPS protocol usage, as well as using the Python library `DNSPython` to query for other information. The usage of the LLMs ChatGPT and Gemini were used to better understand the format of making queries with `DNSPython`, in tandem with the reference material provided by the library’s developers, which were cross-referenced and tested for their specific output. Otherwise they were not used in the data collection, with the special exception of debugging the DNSSEC extraction function when it was giving no response. The research team was alerted to the fact that some ISPs strip the records of any mention of DNS first through LLMs, after which the suggested fixes were tried, and once those failed, additional research on how to avert this was done outside of it, and then implemented. Once the data is extracted, the raw data list is checked for the presence of several different objects. These objects are as follows: HTTPS usage, the presence of HTTPS RR, HTTPS RR + ECH usage, HTTPS RR + DNSSEC usage, HTTPS RR + ECH + DNSSEC usage, and a series of HTTPS RR parameters. The parameters we check for are `SvcPriority Mode` used (either `Alias` or `Service`), `ALPN Usage`, `ipv4hint Usage`, and `ipv6hint Usage`.

For the TLS handshake, what we did was first try to make a handshake with the apex domain, then if that failed, we tried to make a handshake with the `www` subdomain. The socket library of Python was used to make a TCP connection on port 443, with a timeout of 5 seconds. If the handshake was successful with either domain, then a `True` was returned, which then was added to our tally of domains using HTTPS protocol. Should the apex domain connection have failed for any reason, including timeout, then another connection was made using the `www` subdomain, which, if it then failed, returned a `False` that results in no increment of the tally. Because we eventually returned false in the event of a repeated connection failure, there might have been an issue of undercounting with regards to the number of domains using HTTPS protocol.

With all other tallies, our data extraction was dependent on the existence of an HTTPS Resource Record. If the existence of HTTPS RR in a domain has failed to be found, then not only is the HTTPS RR tally not incremented, but so is every searched for object in the data list, aside from the count of HTTPS protocol. In the case of ECH and DNSSEC, this is because we were trying to track their simultaneous usage with HTTPS RR, whereas with the HTTPS parameters, checking for them specifically relied on reading an HTTPS RR, as they are specific to these records. The way we specifically checked the existence of an HTTPS RR was to query the domain using a series of DNS resolvers (`'8.8.8.8'`, `'1.1.1.1'`, `'9.9.9.9'`, `'208.67.222.222'`), once again with 5 second timeout. Both

the apex domain and, if that failed, the `www` subdomain were checked 4 times (in the case of a timeout) via UDP connection for the existence of a HTTPS RR. Should that have succeeded either time, then both the record and the specific domain was returned, resulting in an incremented HTTPS RR tally. Should the `www` subdomain either get no response once, or either timeout or fail to connect for a fourth time, then a Python `none` was returned, resulting in not tally increment. Because of the timeout rule, there exists the possibility that every statistic relying on HTTPS RR is undercounted, but we believe them to at least be somewhat representative, due to their resemblance to previous studies [1].

If the HTTPS RR query succeeded, then an ECH check was made on the returned record. If there was a reference to ECH in the resource record, then a `True` was returned, resulting in the HTTPS RR + ECH tally incrementing. Otherwise a `false` was returned, and no such increment occurred.

A successful HTTPS RR query also returned the targeted domain in addition to the record, the former being used for the DNSSEC query. Using UDP and a series of resolvers (`'1.1.1.1'`, `'8.8.8.8'`), the targeted domain was checked for both the presence of the AD bit and an RR Signature. Should either have been found, then a `True` was returned and the tally is incremented.

In the event that an HTTPS RR was found with both ECH and DNSSEC, then another tally was made for HTTPS RR + ECH + DNSSEC.

Afterwards, the parameters were checked for, all at once. The returned resource record was checked for the `SvcPriority` mode, and if there was the presence of `alpn`, an `ipv4hint`, or an `ipv6hint`. The `SvcPriority` mode was returned, as well as a boolean for each of the other parameters. Appropriate tallies were made depending on that return result.

Once all the results are returned, then the tallies themselves were made. These values were attached to the appropriate domain, and then outputted in a JSON list. Additionally, the tallies were used to calculate the share of each object measured by taking their count and dividing by the number of domains (10,000). These calculations, along with the tallies themselves, were also outputted to a summary JSON file. The output domain lists were then split into further JSON lists, each cataloging the domains containing a specific record and protocol combination.

B. Qualitative Data Collection Approach

Once split records were made, we checked specifically the list of domains that simultaneously used HTTPS RR + ECH + DNSSEC. We were unable to check the other list given time constraints; all lists can be viewed on our GitHub repository in section 5.

Each domain had two types of categorization applied to them: Site Category and Desired Content. Site Category described the broad type of site that a domain embodied. For example, these could have been news sites like The New York Times’s site, or a chat site like Discord. Desired Content described the primary form of content that a user would have

wanted to see from that site. An example would be Text, which described the desired content of sites that share purely textual media, like fan fiction sites.

The categorization process was as such: a researcher goes through the split lists described above. Based on the information they could gather around, they decide the specific Site Category and Desired Content that most closely matches the site. They were forbidden from directly visiting a site, unless its safety could have been proven. That safety process is broken down in the Ethics section. This was for their safety, as directly visiting unknown sites can be risky, potentially housing malevolent content like malware, or feature some form of data spying. The researcher was to look for surrounding information about these sites, be it from another site or the search engine snippet. This method was subject to the personal biases and judgment of the researcher performing the task, as well as the potential for them to be fed incorrect information, so categorization may have varied.

These categories had many edge cases amongst themselves and with each other, where either a site of one category could potentially be described as another category, or situations where a particular category could be considered a subsection of another. In instances where two categories are similar, or that a site could be considered two categories of the same type, the dividing line for Site Category is what category can the site be described as closer towards, whereas Desired Content is decided based on what category is closest to the primary form of content the user consumes on that site. There are instances where sites can be considered of two or more categories simultaneously. A site whose Desired Content is Adult_Entertainment or Pirated_Media can often also be considered a Video_Streaming site, but their desired content is most closely described as the former, and these are what will be driving the site's security incentives the most. We will explain the breakdown of each category, their reason for being a distinct category, and their edge cases and why they do/don't apply in this section. Additionally, the decision to only have two categories with these edge cases, rather than a format that allows more than two classifiers, means that some information on these sites will be lost. This decision was made for the sake of easily reading overall trends.

C. Site Categories

Each Site Category is as follows:

Chat_Site: Refers to any site that is primarily used for communication through messages among users. This can include sites that are broadly referred to as social media, but not exclusively. A site like Discord and Telegram are examples of a Chat_Site. Other traditional social media sites, such as networking sites like LinkedIn, or micro-blogging sites like Twitter or Instagram, are all not examples of a chat site. The reason for this is that these other sites arguably do not have the facilitation of direct interaction as their primary purpose, which creates different security considerations and incentives.

Health_Site: These are sites that are primarily devoted towards physical or mental health in one way or another. They

can be, but not exclusively are, health news aggregators like WebMD, actual hospital sites, health related charities, or a health advice site.

Image_Browser: Describes a site used primarily to browse through sets of photos. This can describe a photo hosting site like Flickr or Imgur, as well as piracy sites for things like manga.

Interest_Site: Sites in this category are ones dedicated to a particular interest, but don't fall under the other categories. These would be things like certain recipe sites or hobby sites, that once again, don't fall under any other category.

News_Site: These are sites that primarily exist to provide news on anything in particular. It can be for political events, industry-specific occurrences, or a hobby, so long as the site is meant to relay what's happening within a subject.

Ref: Describes any site that is used as a reference document. This can mean online encyclopedias, dictionaries, manuals, and documentation.

Service_Site: A site that is used to provide or sell a service. It can be any price, and can occur at any particular location or happen remotely.

Shopping_Site: A site used to buy something that isn't a service or something that causes the site to fall under another category.

Software_Sharing: This is meant to describe sites that exist to share software for free, without a service attached, or at least provides a free non-trial version of the software. This can describe any type of freely distributed software, as well as software that, while having a paid version, has a non-trial free version that is feature complete or near feature complete. So both 7Zip and WinRAR fall under this category, but not WinZip, which forces the user to pay for it after a free trial.

Text_Sharing: Describes sites that exist to share different forms of textual content. That textual content can take multiple forms, such as text being hosted within the webpage itself, or as a document file. Examples of this would be free online library sites, fan fiction sites, or literary piracy sites. The sites for physical libraries fall under the category of service sites.

Unknown_Site: Describes any site the researcher cannot find enough information to categorize, whilst also not being safe enough to visit.

Utility_Site: Sites that perform a specific tool function for a user. These can be to convert files, edit images, or create specific types of files.

Video_Site: These are sites that exist to share videos. This can be in streamed or downloaded form, so long as the site exists mostly to share these videos.

D. Desired Content

Each Desired Content category is as follows:

Advertising: This is the content of a site that exists to advertise something. A traditional example would be the site for a game or movie. It could be for some kind of software or service you cannot directly buy on said site. This category is superseded by any other category that fits the site.

Adult_Entertainment: Describes any kind of sexual content that is interacted with primarily for that inherent trait, be it videos, images, services, or any other format. This category supersedes other categories, like **Pirated_Media** or **File_Sharing**, due to the heavy influence it has on domain security decisions.

Articles: Describes textual content that makes in-tandem use of images or videos. Also stands in for mixed media sites, like recipe sites or tutorial sites with heavy text and video portions.

Communication: This category describes site content that is defined by direct user to user communication. These can be texts or file sharing, so long as it's initiated between one or more users.

Conditional_Software: Software that is shared under a condition. An example of this condition can be to share data, or to pay for it either as a service or as a standalone product.

Database_Info: Content that revolves around information relating to a database item, and its relationship to other items in the database, while also not following under any other category. This includes databases dedicated to databases on movies and music, as well as sites that track the history of item prices in a marketplace.

File_Sharing: Site content that takes the form of shared files that are supposed to be downloaded by another user, and does not fall under another category, like **Pirated media** or **Monetary Transaction**.

Monetary_Transaction: These are any type of site content that induces a monetary transaction of any kind. These can be for shopping sites, service sites, donation sites, financial/asset transfer sites (PayPal, Venmo), or banks.

Piracy_Media: These are any form of content that is shared in the form of piracy, or that is commonly used to facilitate piracy. They can be streamed or downloaded, but they are specifically either pirated copies, or software that is commonly used to obtain pirated media. Once again, sites with this content can fall under other categories, but this supersedes everything except **Adult_Entertainment**, due to how it affects privacy and security incentives for a domain.

Text: Any content that is mostly text, but doesn't fall into the other categories. In this case, this would be text that is outputted by the webpage.

Unknown_Content: Describes any content the researcher cannot find enough information to categorize, whilst it is also not safe enough to visit the hosting site.

Utility_Software: The content of any site that's meant to perform some free practical utility, without also being a part of the other categories. These include things like file conversion or looking up one's own IP address. This can also take various forms, like an on-webpage utility, or as a downloaded executable file.

V. ETHICS

In order to collect and measure data ethically, we minimize the workload on the domains we're fetching, which means that we are asking for measured parameters and usage details only from the domains without the extra data we don't need. In addition, we respond to any requests to stop measuring

the data. The reason minimizing workload is ethical is that it can maximize benefits while minimizing harm to affected systems and users [11], since sending too many requests to a domain can cause server overload, bandwidth consumption, or exhaustion of computing resources.

The domains surveyed may also have malevolent content, such as malware. To mitigate danger to the data collector, several precautions are taken. The Tranco list that is generated is specifically configured to exclude domains that Google Safe Browsing lists as dangerous. Furthermore, researchers avoided direct site visits unless necessary and verified site safety prior to access.

VI. RESULTS

These are the results of the data extraction process and qualitative data collection. All share percentages calculated are rounded to the second decimal place. While the domains visited are the top domains of 2025, the actual data extracted is from March 13 2026, and is thus reflective of the DNS environment of that particular day. The raw extracted data, calculation, and data extraction tools can be found in the our GitHub repository (<https://github.com/RhoAI/CS578-Project-Chain-Link-Security-Data-Files>).

Tables 1-3 were made in Microsoft Excel and converted to $\LaTeX$, while Figure 1 was made using the LLM Gemini, which took an Excel pivot table that we created as input to produce a heatmap. The values for every table, except Figure 1, was calculated from the raw results of the data extraction step. Figure 1 itself was calculated from a table made using the values collected during the qualitative data collection

Table 1 showcases the count and percentage of the domains that use HTTPS, HTTPS RR, HTTPS RR with ECH, HTTPS RR with DNSSEC, and HTTPS RR with both protocol sets simultaneously, from the top 10000 domains.

For Table 2, it describes the share each HTTPS RR security protocol configuration (HTTPS RR+ ECH/ HTTPS RR + DNSSEC /HTTPS RR + ECH + DNSSEC) as part of the total number of HTTPS RR domains.

Table 3 shows the amount and percentage of the domains that contain different HTTPS parameters among all domains, as well as the domains using HTTPS RR.

Figure 1 is a heatmap that describes the cross-section of different types of Site Category and Desired Content, for domains that are simultaneously using HTTPS RR + ECH + DNSSEC.

VII. DISCUSSION

The calculations from our data extraction largely saw a continuation of trends seen in previous studies. The share of domains using HTTPS RR in 2026 is greater than that observed in 2024 [1] or 2023 [2], at around 30%. However, the share in this study is not exactly comparable to the shares in the other ones, as this share reflects the percentage against top 10,000 domains, whereas the discussed observations either dealt with 1 million domains, or all domains of a particular type, so the 10,000 domains as a whole are substantially more

TABLE I
COUNTS AND SHARES OF ALL HTTPS CONFIGURATIONS AGAINST ALL DOMAINS

Record/Protocol	HTTPS Usage	HTTPS RR	HTTPS RR + ECH	HTTPS RR + DNSSEC	HTTPS RR + ECH + DNSSEC
Count	8641	3126	773	745	71
Share(%)	86.41	31.26	7.73	7.45	0.71

TABLE II
SHARES OF HTTPS RR CONFIGURATIONS AGAINST TOTAL HTTPS RR DOMAINS

Protocol	HTTPS RR + ECH	HTTPS RR +DNSSEC	HTTPS RR + ECH + DNSSEC
Share(%) of HTTPS RR Domains	24.73	23.83	2.27

TABLE III
COUNTS AND SHARES OF HTTPS RR PARAMETERS AGAINST ALL DOMAINS AND HTTPS RR DOMAINS

HTTPS params	Alias mode	Service mode	alpn	ivp4hint	ivp6hint
Share(%) vs all domains	0.04	31.22	31.08	27.49	23.27
Share(%) vs HTTPS RR	0.13	99.87	99.42	87.94	74.44

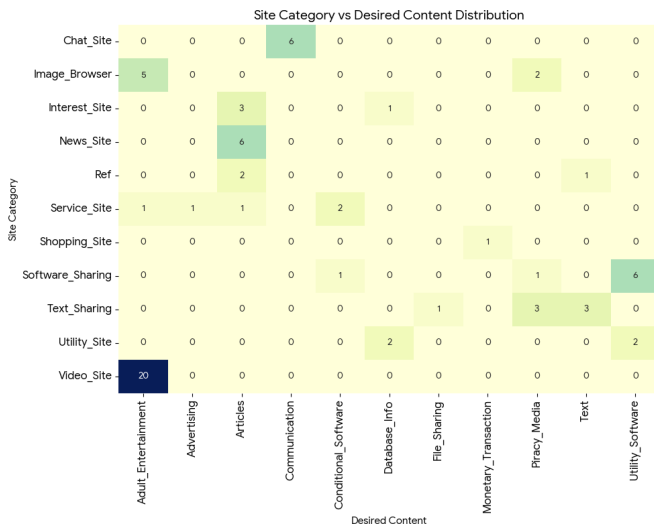


Fig. 1. Heatmap of Site Category and Desired Content

visited than the vast majority of the other domain sets, which is likely to have more resources to support HTTPS RR than less popular domains, making a direct comparison meaningless. The simultaneous usage of HTTPS + ECH is much higher than the larger dataset from 2023 in terms of just pure tally [2], which is significant, but also expected as the DNS record HTTPS RR was just published. This value is also in line with the general amount of ECH within TLS traffic in 2022, both of which are around 7% [12], suggesting growth of ECH use only in the context of HTTPS RR. The usage of HTTPS RR + DNSSEC is around the same percentage as examined in the previous studies, as are the parameter shares. So largely, either comparisons to the values from older examinations cannot directly be made, or are in line with the previous values. This means that the data will have to be taken on its own.

Taking that into account, HTTPS takes up the majority of protocol usage for the top level domains. In turn, HTTPS Resource Record usage is a serious chunk of both the total

top 10,000 domains, as well as HTTPS domains in general, accounting for roughly a third of them in both instances. Both HTTPS RR + ECH and HTTPS RR + DNSSEC take up a small, but not insignificant portion of domains at around 7%. Simultaneous usage of HTTPS RR + ECH + DNSSEC was, however, incredibly small. It accounted for less than 1% of total domains examined, total HTTPS domains, and was barely much higher as a share of HTTPS RR domains. HTTPS RR + ECH and HTTPS RR + DNSSEC, despite their security, seem relegated to a niche, and simultaneous usage of the records and both security suites are rather rare, in the top layer of domains where resources are likely highest.

There could be a number of reasons for this. Encrypted Client Hello (ECH) improves privacy in TLS, but its implementation carries with it disadvantages with the advantages. It requires a lot of different components to align in the internet infrastructure for it to successfully work. This includes things like DNS configuration, TLS implementation and server side support. The encryption key rollovers, as discussed before, are an especially worrisome aspect of that, requiring both client and server to manage the keys every 1 to 2 hours in order to avoid connection failure [1]. All these components require work to be done individually which means increased maintenance managing them. Some organizations may deem it unnecessary unless there is a strong incentive to opt in.

Another factor affecting ECH implementation is its impact on content filtering and network monitoring. When ECH is being used, it encrypts the Server Name Indication (SNI) during the handshake. While ECH advances user privacy by concealing sensitive metadata within the TLS handshake, it complicates content filtering solutions depending on this information visibility [13].

Moving beyond, DNSSEC adoption itself remains low across the internet. One of the reasons for this is complexity of deployment and maintenance. Implementing this requires careful management of keys and rollovers, which can cause DNS failures if they are not handled properly. A single mistake causes the domain to crash for all validators. These can cause

major shutdowns for high profile websites, and some of these operators are small, lacking the cryptology experts to resolve events like these. Maintenance costs are also complicated, as handling these cryptology protocols can increase upkeep [14].

Secondly, DNSSEC does not really protect the threat that most clients care about. With DNSSEC, it only ensures authenticity and integrity of DNS records on the server side. Confidentiality is not provided for a client, so their information can still be accessed. This forces DNSSEC usage into the specific sector of high-security domains, like banks and government sites [14].

Adoption has also been affected by the emergence of alternative DNS security technologies like DoH and, to a lesser extent, DoT [14] [15]. DoH and DoT encrypt DNS queries between clients and resolvers, providing the privacy protection that most users look for, hence the opt-in for domains. Even though the technologies take care of different security issues, the fact that these alternatives are there, are available, and are easier to use reduces the chances that these domains would see the need to use DNSSEC.

What DoH does not give you, that DNSSEC does, is end to end validation without trusting a third party resolver like Cloudflare, and protection against a malicious or compromised public resolver that may be lying. Despite these, most users and organizations have decided they are fine trusting third-party resolvers. So application of DNSSEC is highly specific and niche, mostly seeing DoH and some optional DNSSEC validation to solve most DNS security problems that arise [14].

The niche uses of these security protocols beg the question which domains do use them, especially in the highly specific application of HTTPS RR in tandem with ECH and DNSSEC. This situation is partially illuminated through examination of the qualitative data in Figure 1. HTTPS RR + ECH + DNSSEC sees particularly high usage among a specific set of domains. Sites that feature adult entertainment-especially in the form of video and image browsing content-are common, as are sites related to piracy. News and interest sites with heavily mixed media articles, as well as sites that share files and software are also among the frequent clients.

While the exact motivations for the adoption of the security protocol suite can only be known by the domain holder, the incentives that can push these domains into their use are clear and can be easily extrapolated. In the case of sites with adult entertainment and piracy related content, both these subjects can be of dubious legality in certain areas, and the former can also be taboo. Client association with either is therefore undesirable, incentivizing the use of ECH's ability to hide SNI. The incentives driving the other common clients, with regards to ECH, are harder to evaluate, but can still be estimated. Software and file sharing sites, specifically the communities the development teams of these sites come from, tend to value a strong degree of confidentiality. Encouraging the privacy of clients who use their software or files is seen as a positive, and can build to their own positive reputation, incentivizing the usage of tools like ECH to support that. With regards to news and interest sites with mixed media articles, it can be

a little difficult to see what the larger benefit is. New sites may value a strong degree of reader privacy, so they aren't targeted in anyway, or least they value the idea of it, most of the news and interest sites that use HTTPS RR + ECH + DNSSEC are hobby and science sites, which don't generate the same degree of controversy. Still, many of these sites feature user accounts, which themselves carry personal information, incentivizing some degree of privacy action.

Among these sites, the usage of DNSSEC is much clearer. Many make use of large files that would require heavy cache use in a Content Delivery Network (CDN): video files for streaming sites, picture files for image browsers and articles. That means that a lot of these sites are large potential targets for cache poisoning [16]. In the case of software and file sharing sites, their reliance on user downloading files make them targets of spoofing attacks, where adversaries can take advantage of the userbase's expectations to download malicious files or give confidential information away. Both these attack types are protected against through the DNSSEC's content authentication, encouraging its use among these sites.

VIII. CONCLUSION

The larger environment for HTTPS RR deployment looks like a logical continuation of what came before. HTTPS RR is utilized by a significant chunk of popular domains, but it is difficult to tell if that adoption is growing. ECH adoption in the context of this new record is up, but overall adoption is largely the same. A similar situation is found with the adoption of DNSSEC, which seems in line with what came before.

Despite the promise of simultaneous usage of HTTPS RR with ECH and DNSSEC, the combination is plagued with maintenance difficulties, limiting its use outside of heavily incentivized environments. Furthermore, some of its protections are provided by easier to use alternatives, making the suite overall unattractive to most popular domains. Fittingly, its adoption is minuscule compared to other deployments of HTTPS RR.

That being said, the combination of HTTPS RR with either or both of the security protocol suites have their niche use-cases. ECH is especially desirable in scenarios where user confidentiality is valued. DNSSEC can be useful in domains that are large targets for different DNS attacks. And as seen in the heatmap, sometimes these incentives can overlap.

HTTPS RR combined with both protocol sets has few willing clients, but there does exist a niche of domains with a cross-section of the incentives for both DNSSEC and ECH. These generally are domains that are encouraged to strongly protect user confidentiality, while also using the cache heavily enough to be vulnerable to poisoning attacks, or a structure that can be exploited with a DNS spoof attack. These situations thus call for the tandem usage of DNSSEC and ECH for protection.

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